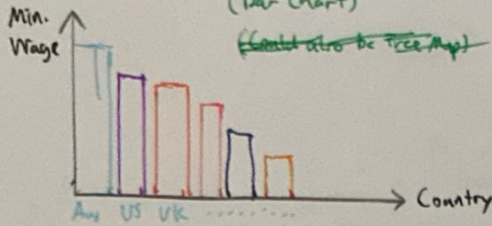


IDEAS

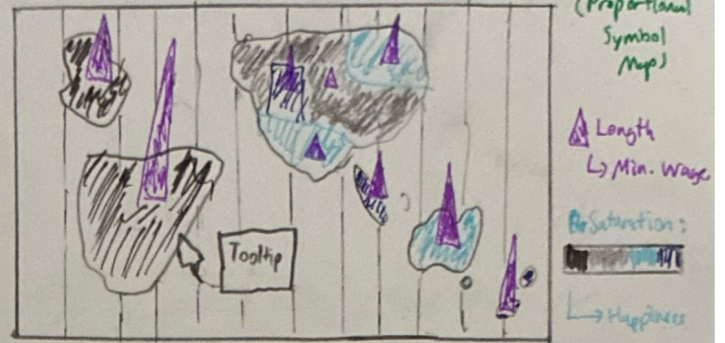
Quality of Life - Global

Randal Wallace
19 September 2025
Sheet 2
5 - Design Sheets (Assignment 2)

① CPI-Adjusted Minimum Wage (Bar Chart)



② Minimum Wage vs Happiness/Generosity Levels (Proportional Symbol Map)

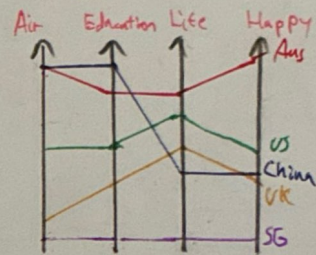


③ Each Country's Metrics - Comparison (Parallel Coordinate Plot)

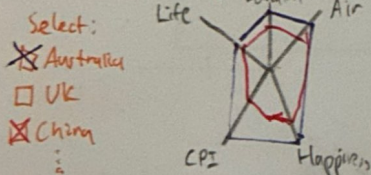
Table:

	Metric (Must be normalised)				
Country	Air Quality	Education	Life Expectancy	Happiness Levels	
Aus.	85	95	90	80	
US.	95	90	80	90	
UK.	65	100	80	70	
SG.	40	100	80	10	
China.	50	90	70	100	

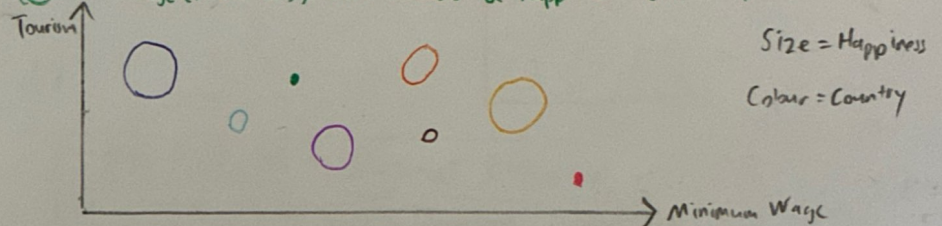
Chart:



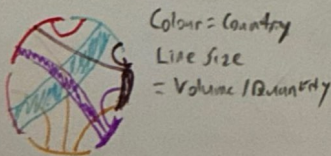
④ Individual Country Metrics (Radar/Spider Chart)



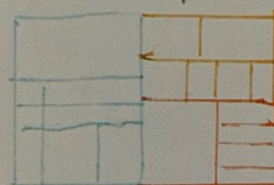
⑤ Min-Wage (Adjusted), Tourism Levels & Happiness (Bubble plot)



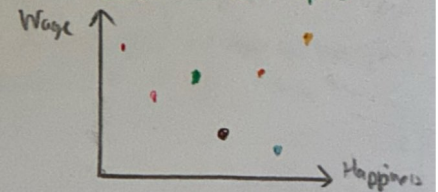
⑥ Chord Diagram



⑦ CPI - Adjusted Min Wage (Tree map)



⑧ Same Happiness vs Wage Resolution (Scatterplot)



FILTER

X ⑥ - Wrong type of data
- Too hard to program

X ⑧ - Does not highlight trends well

✓ ① - Plain and simple
- Easy to code
- Good introduction to narrative flow

✓ ② - Excellent interactivity
- Harder requirement for assignment

✓ ③ - Can highlight global trends

✓ ④ - Allows small-scale comparison
- Interactive

✓ ⑤ - 3 trends at once
- Intuitive design

CATEGORISE

Qualitative / Categorical

- Country
- Region
- Education (Primary / Secondary)

Quantitative

- Education (Years)
- Minimum Wage
- Working Hours
- Consumer Prices
- Air Quality
- Happiness Levels
- Tourism Levels
- Life Expectancy

COMBINE & REFINES

① chosen over ⑦

- ① is easier to program
- Data for region may need to be hard-coded
- ⑦ may be too overstimulating with like 75 regions

QUESTIONS

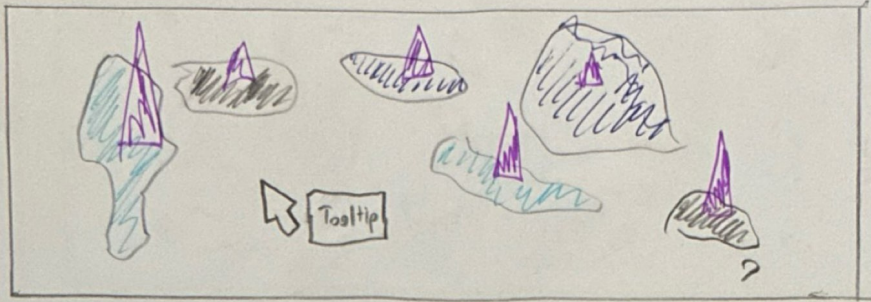
- Is region a possible thing to query using Javascript & Vega-lite? (Now category)
- How will I adjust monetary attributes
- Accuracy of data - times are not all the same

ABOUT

Quality of Life - How does your country fair?

TOP OF WEBSITE

②

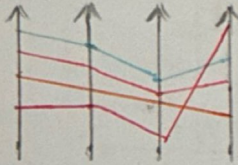


Length: Minimum Wage

Colour Gradient: Happiness Level

TEXT TEXT TEXT TEXT

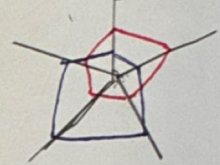
③



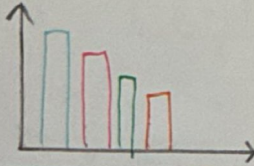
④

Select Countries:

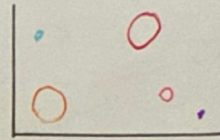
☒ A
☐ B
☒ C
☐ D



①



⑤



FOCUS

Narrative Flow

- Charts organised in a way where the viewer can be introduced to the topic in an intuitive way

Interactivity and focus on chosen countries

- Allows user to personalise the dashboard
- Interactivity allows user's own curiosity to make them explore more statistics.

DISCUSSION

OPERATIONS

JavaScript & VegaLite

- Need to download world map with countries
- Tooltips must be manually programmed

Scrolling is a feature

- Should have a good amount of visualisations, whitespace and text take up a standard desktop size at all times.
- CSS boxes ☹️

Adjusting Data

normalize

- May need to adjust some monetary attributes to ~~per~~ per capita or purchasing power.
- Older data should be excluded

① - A bit TOO simple.... will keep it at the back of my mind for now...

④ - Could have a very user-friendly UI to encourage viewers to play with it.
- CSS is your friend.

③ - Hopefully we arrange the vertical arrows (quantitative attributes) in a way where there is at least some obvious trend.

Randal Wallace
19 September 2025
Sheet 2

5- Perlin Sheet (A2)

LAYOUT

TOP
HALF
OF
WEBPAGE

SCROLL

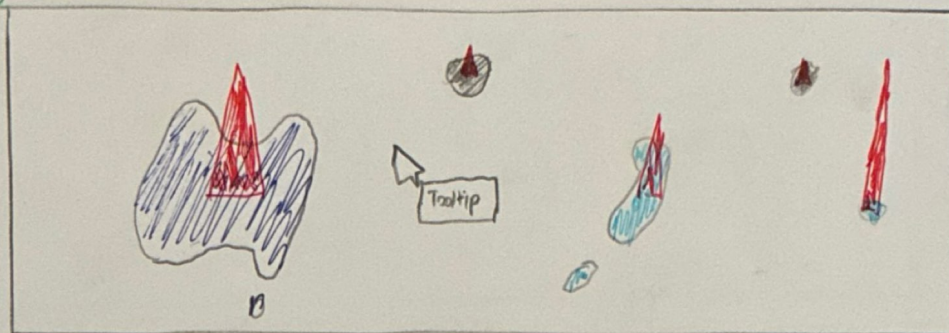
SCROLL

SCROLL

BOTTOM
HALF

"QUALITY OF LIFE AROUND THE WORLD"

②



Legend:

▲ Length → Min wage

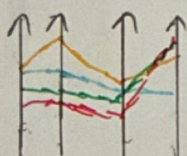
Colour Gradient

→ Happiness Level

[May add other buttons here]

TEXTBOX

③



TEXT
[Overall trends
globally]



④

Select Countries

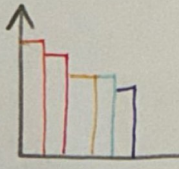
☒ A (Toggle)

☐ B

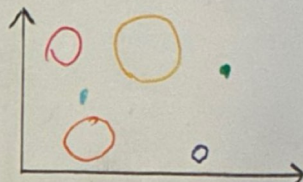
☐ C

WHITESPACE (or some cool animation/design)

①



⑤



WHITESPACE

METADATA: [Author, Source, etc.]

FOCUS

Whitespaces

- Make narrative flow and visual hierarchy VERY obvious
- Use whitespace strategically

Space Textboxes and Legends/Buttons

Strategically

- Just make the website easy for someone to use even if they are illiterate!

OPERATIONS

Tooltips for ⑤

- Implement something like:

Country Name:
Happiness Level:
Min Wage: ← smaller font
Tourism Level: ← smaller font

- Adjust font sizes, type and colour using CSS.

Javascript for ④

- Will have to figure out how to program the checklist-style box to show only countries he are interested in.
- Hopefully VegaLite makes this easy...

DISCUSSION

- ② - Maybe using Δ length on top of countries is not very intuitive and would not discriminate smaller discrepancies between countries.
- Very hard to compare lengths of countries if they are separated by a lot of space geographically

- ⑤ - I wonder if a bubble chart is really the best way to show these correlations

Randal Wallace

20 September 2025

Sheet 3

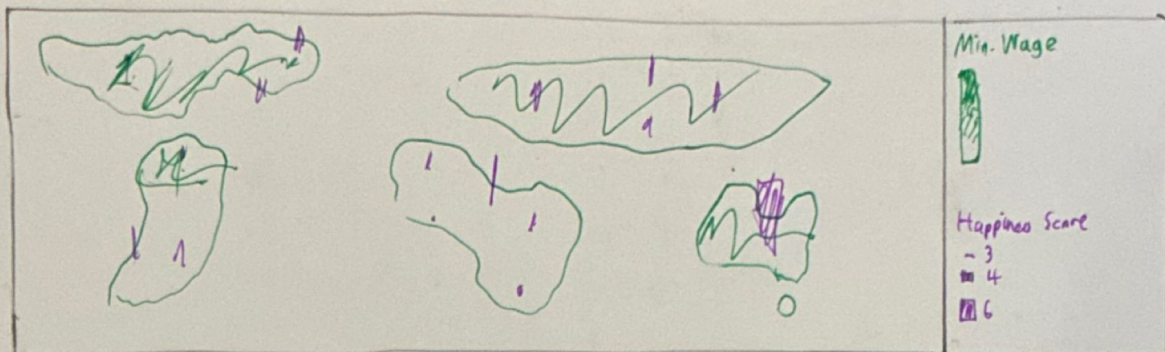
5-Design Sheets (A2)

IDEAS

AYOU

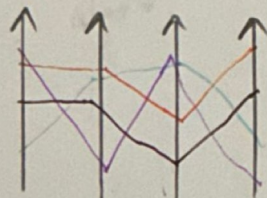
QUALITY OF LIFE AROUND THE WORLD

②



TEXT

③



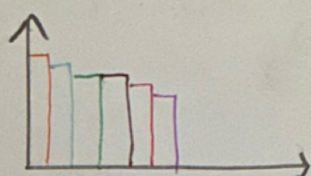
④



Select Country
☐ A (Toggle)
☐ B
☐ ...

W H T T E S P A C E

①



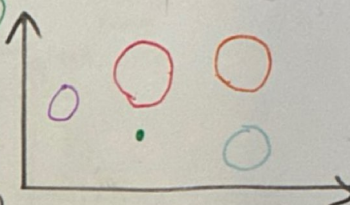
Sort

Filter

Region??

could also affect

⑤



METADATA

Focus + OPERATIONS

Sorting by region and filter by region

- Program JavaScript buttons
- Can filter country graphs
- ① and ⑤ by continent/region
- Lookup continent on JSON file

Ensuring world map stretches to fit webpage screen

- May need to use Flexboxes
- Or adjust manually to big dimension

Zooming into Map

- If possible, should be programmed.
- I cannot see small countries on 1440p monitor

DISCUSSION

Other potential filters?

- Could make all charts have a filter?
- Maybe toggle
- Would make ④ less interactive...

⑤ - still unsure if this is the best chart

Randal Wallace

23 September 2025

Sheet 4

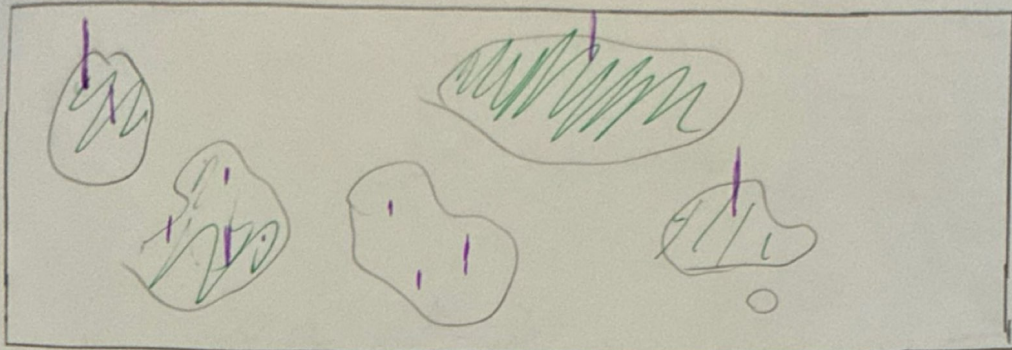
5-Design Sheets (A2)

LAYOUT

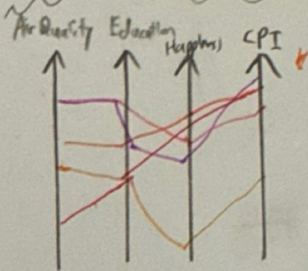
QUALITY OF LIFE AROUND THE WORLD

Ranlal Wallace
24 September 2025
Sheet 5
5-Design Sketch (A2)

②



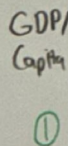
③



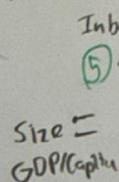
Filter by Region:

- Asia-Pacific
- North America
- ...

Mention any normalizations and year of data



①

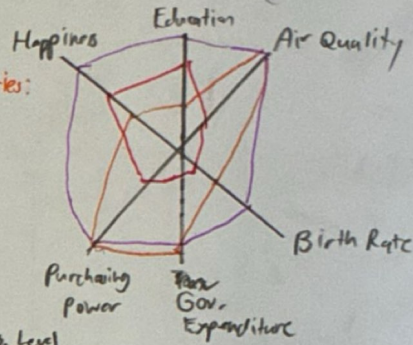


⑤

Select Countries:

- A
- B
- C
- ...

④



⑤

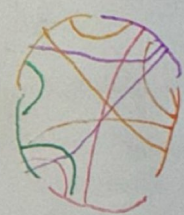
Tooltip

Size = Healthcare Structures

Government Expenditure

- Include:
- Birth rate
 - Overall population
 - Fertility rate
 - Language

⑥



Colour = Region

Arrow = Direction of Travel

METADATA

FOCUS

- Whitespace
- Visual Hierarchy
- Intuitive correlation between elements in visualisations
- Tooltips should go into more detail
- Filtering by region
- Buttons and interactivity

OPERATIONS

Added more attributes

- Categorical:
- Language
 - Outbound tourism
- Quantitative:
- GDP/capita
 - Government Expenditure %
 - Mortality/Healthcare
 - ~~Outbound tourism flows~~

Chord Diagram ⑥

- Coloured by region
- Size of arrows is irrelevant
- Must filter country JSON map file by continent/region

DETAILS

Filter by Region

- Not applicable on: ②, ⑥, ④, ③

All region colours must be consistent!

Ensure data is clean

- Normalise if necessary
- May need log scales and time period
- Changes should be mentioned on dashboard
- Use most recent year data